Year 10 Higher

REVISION PACK

Graphs 1

Name: _____

Class: _____

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: a missing coordinate

A straight line passes through the points (1, 4) and (5, k). The gradient of the line is 2. Work out the value of k.

Step 1	Gradient formula	$\frac{k-4}{5-1}=2$
Step 2	Simplify the bottom	$\frac{k-4}{4}=2$
Step 3	Multiply by 4	$k-4=8$
Step 4	Add 4	$k=12$

Practice: a missing coordinate

1.1 A straight line passes through the points (3, 2) and $(7, k)$. The gradient of the line is 4. Work out the value of k.

1.2 A straight line passes through the points (1, 9) and $(4, k)$. The gradient of the line is −2. Work out the value of k.

1.3 A straight line passes through the points (−2, 5) and $(3, k)$. The gradient of the line is 3. Work out the value of k.

1.4 A straight line passes through the points (0, −3) and $(4, k)$. The gradient of the line is 2. Work out the value of k.

1.5 A straight line passes through the points (2, 1) and $(5, k)$. The gradient of the line is −3. Work out the value of k.

1.6 A straight line passes through the points (−1, 6) and $(3, k)$. The gradient of the line is 2. Work out the value of k.

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: the equation of a line through two points

Find an equation of the straight line that passes through the points (−2, 9) and (2, −3).

Step 1	Gradient	$m = \dfrac{-3-9}{2-(-2)} = -3$
Step 2	Put in a point	$9 = -3 \times (-2) + c$
Step 3	Solve	$c = 3$
Step 4	Answer	$y = -3x + 3$

Practice: the equation of a line through two points

1.7 Find an equation of the straight line that passes through the points (2, 7) and (5, 19).

1.8 Find an equation of the straight line that passes through the points (−1, −1) and (3, 11).

1.9 Find an equation of the straight line that passes through the points (−1, 7) and (2, −2).

1.10 Find an equation of the straight line that passes through the points (1, 6) and (4, −6).

1.11 Find an equation of the straight line that passes through the points (−3, 10) and (1, 2).

1.12 Find an equation of the straight line that passes through the points (0, −4) and (2, 6).